The empirical recurrence for OEIS A283861: recovering a conjecture that is not written down

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Abstract

OEIS A283861 records "Empirical recurrence of order 74", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 147$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 74, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 74 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283861 is "Number of 5Xn 0..1 arrays with no 1 equal to more than three of its horizontal, diagonal and antidiagonal neighbors.". It has offset 1 and begins

$$32, 1024, 19776, 448490, 10741381, 247708452, 5724272337, 133133661082, \ \ldots$$

The entry states:

> Empirical recurrence of order 74 (see link above)

As of the "Last modified" line on the live entry (Mar 17 2017 12:04:50, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 147$ of the 1056, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 147$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 294$ exact terms of the model returns order 74 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{74} c_i\, a(n-i),$$

c_1	14	c_{20}	6287984113513	c_{39}	−62377129033094893	c_{58}	1556051573660422
c_2	86	c_{21}	15035466218915	c_{40}	−35015823619913444	c_{59}	17444693106614215
c_3	2675	c_{22}	−29231226815987	c_{41}	31165648692031793	c_{60}	16578663367860121
c_4	7366	c_{23}	−30218237751565	c_{42}	53873718006609945	c_{61}	10001478903293153
c_5	9187	c_{24}	66913014944865	c_{43}	−13387980291866877	c_{62}	6785590066430941
c_6	−372998	c_{25}	−68150110435621	c_{44}	−27352984118524539	c_{63}	3637331201063691
c_7	−3668317	c_{26}	−127107817882018	c_{45}	29794290789366156	c_{64}	2413295256940129
c_8	−15244255	c_{27}	178858202054925	c_{46}	−26595725897452661	c_{65}	751718943020690
c_9	−32044000	c_{28}	−460935328496070	c_{47}	117394428180488266	c_{66}	347201607393958
c_{10}	57200116	c_{29}	−709967776623697	c_{48}	−13761889044836587	c_{67}	126105465326786
c_{11}	860643565	c_{30}	2510824208010939	c_{49}	−25658601789096768	c_{68}	1500056624952
c_{12}	2448891047	c_{31}	1324939008126784	c_{50}	−12260130214848191	c_{69}	700115434752
c_{13}	−720409095	c_{32}	−7994426753716621	c_{51}	17745334352617353	c_{70}	−1153674458136
c_{14}	−18569710668	c_{33}	3049429606501294	c_{52}	17090785758566613	c_{71}	−46534915592
c_{15}	−72979395112	c_{34}	17316315194060964	c_{53}	6195268680833883	c_{72}	9844372464
c_{16}	−24902329363	c_{35}	−9662322112837158	c_{54}	5901968677641244	c_{73}	236767680
c_{17}	528199520901	c_{36}	−28066608282537607	c_{55}	−37670814220299272	c_{74}	−493056
c_{18}	−116254751440	c_{37}	25641671611725109	c_{56}	44253965824453705		
c_{19}	−2739700486338	c_{38}	61563193686518328	c_{57}	36934800573461002		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 74, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $74+4$ terms removed returns order 74 as well, so $\deg q_{\min} = 74 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $74+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 74 that this sequence satisfies — and that family turns out to have exactly one member.

References

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